Does chain-of-thought still protect against J-space ablation when problems get hard?

CS 2881 Homework 0 — July 31, 2026 — code: `github.com/pgrindehollevik-harvard/cs2881`

1 Hypothesis (pre-registered before any experiment; tag `prereg-v1`)

Gurnee et al. (2026) report that ablating a model's J-space — zeroing the residual stream's projection onto the k=10 most-active Jacobian-lens directions at every position across a mid-layer band — collapses internal multi-step reasoning while sparing shallow capabilities, and that GSM8K solved *with* chain of thought is far more robust to this ablation than the same problems answered directly. We tested whether that protection persists as problems get harder, on `Qwen/Qwen3-4B`. **H1** (replication): on GSM8K the random-differenced protection $P_D = [R(\text{CoT}, J) - R(\text{dir}, J)] - [R(\text{CoT}, \text{rand}) - R(\text{dir}, \text{rand})]$ is $\geq +0.15$, where R = accuracy retention (ablated/clean). **H2** (extension): protection *erodes* with difficulty (GSM8K → MATH-500 → AIME; within MATH-500, levels 1→5), because externalization is bandwidth-limited: each step of a harder problem itself needs more un-externalized workspace. Falsification criteria, floor rules, sample sizes, and a hypothesis-blind calibration gate were frozen in `docs/PREREGISTRATION.md` before any run; all deviations are logged in `docs/DEVIATIONS.md`.

2 Experiment design

Two prompting modes (CoT vs. direct-answer, both non-thinking mode, matched sampling) × three conditions (clean / J-space ablation / *per-slot norm-matched* random-direction ablation) × three datasets (GSM8K n=150; MATH-500 stratified 30/level; AIME 2024+2025, all 60). The random control removes an exactly equal-norm component along random directions at the same (position, layer), so anything surviving the P_D subtraction is J-space-specific rather than generic damage. Before the main grid may run, a hypothesis-blind stop-gate requires: a significant positive-control drop under J-ablation (exact McNemar), $\geq 2\times$ the matched-random drop, output degeneration $< 10\%$, and ordinary next-token prediction on held-out wikitext surviving (> 0.80 top-1 agreement with clean) — the paper's selectivity signature. Strength (layer band, k) is calibrated *only* on these criteria, never on CoT-vs-direct gaps.

3 Experimental details

No ablation code exists in the companion repo; we implemented it as forward hooks: per (position, band-layer), readout logits = lm_head(rmsnorm($J_\ell h$)), top-10 token ids (recomputed per position and layer, ids in the clean model's top-10 next-token predictions exempted via a parallel clean stream over the same tokens), directions $v_t = J_\ell^\top(\gamma \odot W_U[t])$, joint projection removed by regularized batched solve. Ablation applies to prompt and every generated token. Pre-fitted lens: `neuronpedia/jacobian-lens` (fit on this exact checkpoint); it beats a logit-lens baseline at ranking silent multi-hop intermediates (pass@5 0.41 vs 0.37; pass@1 0.20 vs 0.19), validating it before use. Percent-depth transfer of the paper's Sonnet band gives candidates $\{14\text{–}24, 14\text{–}31\}$ of 36 blocks (light = 14–19). Compute: one M4 Pro (48 GB, MPS), 19.3 tok/s clean, 6.2–6.6 tok/s ablated (dual streams + per-layer readouts). Runs are chunk-resumable (64-token persistence): macOS update preparation repeatedly SIGKILLed the process (90 s lifetimes at the worst), which cost wall-clock but, by construction, no data. Grading: `math-verify` on the last `\boxed{}`; artifact classes (no-answer / truncation / degeneration / wrong-answer) tracked separately. 7 unit tests certify the intervention (identity when disabled, projections verifiably removed, exemption honored, per-slot norm match, seeded determinism, chunked = unchunked).

4 Results

Positive-control calibration (order-ops fallback; clean multihop was 0.42, far from ceiling — itself a small-model datum). All four pre-registered rungs *failed* the stop-gate, so the main grid did not run on the off-the-shelf lens:

Rung	J drop (pts)	random drop	wikitext top-1 (J / rand)	gate
14–24, k=10	−16.4 (p=.004)	−9.1	0.62 / 0.79	fail
14–31, k=10	−18.2	−9.1	0.58 / 0.73	fail
14–24, k=5	−14.5	−5.4	0.61 / 0.82	fail
14–19, k=10 (light)	−16.4 (p=.012)	0.0	0.66 / 0.84	fail

The J-directions are causally real on Qwen3-4B — in the light band, norm-matched random damage costs *nothing* while J-ablation costs 16 points, with only 3% degeneration — but they are not *selective*: 34–42% of ordinary next-token predictions change (paper: "only mildly perturbed" on Sonnet 4.5). Per the pre-registered contingency (the stock lens targets the final layer; the paper's recipe targets the penultimate layer to avoid last-block artifacts), we refit a penultimate-target lens and re-ran the ladder once.

5 Analysis